

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

7th September 2016

Subject CT6 – Statistical Methods

Time allowed: Three Hours (10.30 – 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** You work for an insurance company ABC which is considering buying a reinsurance policy under which a reinsurer will pay the excess, if any, over Rs 5000 for each individual claim made in the portfolio. A reinsurance company, Expert RE, has offered to provide cover for the whole portfolio for the next year and quoted annual premium of Rs 1600 for the same. You have to decide if the company should go ahead to purchase the reinsurance cover given that there are 1000 policies in the portfolio and ABC expects 20% of these policies will make a claim next year.

You may assume the loss amount for each claim follows an exponential distribution as having a mean and standard deviation of Rs 1000.

- i) Calculate the expected total claim amount ceded to Expert Re over the whole portfolio (5)
 - ii) Based on your answer to (i) comment whether ABC should reinsure with Expert RE. (2)
- [7]

- Q. 2)** The last twelve claims under a particular class of insurance policy were:

2000, 1350, 790, 2500, 825, 942, 77, 1780, 450, 4200, 1525, 3100

- i) Assuming that the claims came from a lognormal distribution with parameters μ and σ , use the method of maximum likelihood to estimate the parameters based on the observed data. (6)
 - ii) Use the method of percentiles (based on the 25th and 75th percentiles) to estimate parameters c and γ , assuming that the claims come from a Weibull (c, γ) distribution. (5)
 - iii) If the insurance company takes out reinsurance cover with an individual excess of loss of 2500 estimate the percentage of claims that will involve the reinsurer under (i) and (ii) above. (3)
- [14]

- Q. 3)** For a certain portfolio of insurance policies, for which it may be assumed that the claims are fully run off by the end of development year 3.

The cost of claims incurred each year is given in the table below:

Development Year

Accident Year	0	1	2	3
2012	3000	1850	720	350
2013	3750	2200	925	
2014	4915	2922		
2015	6200			

The cumulative number of reported claims is shown in the table below:

Development Year				
Accident Year	0	1	2	3
2012	170	250	265	290
2013	195	261	292	
2014	249	310		
2015	272			

- i) Given that the total claims paid to date is 15,000 calculate the outstanding claims reserve for this cohort using the average cost per claim method. (8)
 - ii) State the assumptions that underlie your result. (1)
- [9]

Q. 4) The time taken by employees to pick up a telephone call after the first ring in the direct marketing team of an insurance firm is assumed to follow an exponential distribution with parameter β . 25 calls are monitored at random and the average response time is calculated as 2.372 seconds.

- i) Find out the conjugate prior distribution for β . (2)
 - ii) Assuming that the prior distribution for β has mean 0.450 and standard deviation 0.321, derive the posterior distribution of β and calculate the Bayesian estimator of β under quadratic loss. (5)
 - iii) A further 75 calls are monitored and the recalculated average response time of 2.272 seconds is observed. Calculate the Bayesian estimator of θ under quadratic loss using all the data collected. (2)
- [9]

Q. 5) An insurance company provides insurance against a certain natural calamity. Under the policy the insurer would reimburse the cost of the first three calamities in a year. The insurer models the loss from a single natural calamity as following a Pareto distribution with mean 1 crore and standard deviation 1.5 crore.

The insurer is considering taking out reinsurance for the policy whereby the reinsurer will cover the excess, if any, of the total cost over Rs 3 crore per calamity. The reinsurer has quoted the premiums for two different types of cover as follows:

R1: claims over Rs 3 crore on the first disaster only, with premium charged being 10 % more than the expected cost to the reinsurer (ignore expenses)

R2: claims over Rs 3 crore on each of the first two disasters with premium equal to twice of that charged under R1

The insurer needs to decide whether to opt for cover levels R1, R2 or not to reinsure

- i) Determine the expected total loss per individual calamity, with and without reinsurance. (6)
- ii) Complete the decision matrix below based on the insurers net cost after reinsurance :

	A ₀ (no calamity)	A ₁ (1 calamity)	A ₂ (2 calamity)	A ₃ (3 calamity)
No reinsurance				
R1				
R2				

(6)
[12]

- Q. 6)** A General Insurer believes that the probability of claim of its Marine Insurance contracts depend on age of the voyage (max insured age 10 years) (the linear predictor is given by $\eta = \alpha + \beta X$). The claims follow Binomial (n, μ) distribution.

Probability of Claim	Age of Voyage
0.25	4
0.40	8

- i) Using a suitable link function, find out α and β . (9)
- ii) Find out the maximum and minimum probability which can be attained by the above GLM model. (2)

[11]

- Q. 7)** i) The autocorrelation function of a time series shows coefficients significantly different from zero at lags 1 through 4. The partial autocorrelation function shows one spike and monotonically increases to zero as lags length increases. If the time series is model by ARMA (p, q) then state values of p and q . (1)
- ii) State the definitions of four non-linear time series models. (3)
- iii) If X_t follows MA(1) and $Y_t = 0.6 + 0.3t + X_t$ then prove that the standard deviation of first difference of Y_t will be higher than that of X_t . (6)

[10]

- Q. 8)** A reinsurance company offers catastrophic reinsurance to various insurers. The expected catastrophic claim rate per year is 0.3 and follows a Poisson process. The claim amount follows the following probability distribution and are independent to each other:

Claim Amount (in INR lacs)	Probability
60	0.40
100	0.35
164	0.25

The reinsurance company charges the continuous premium from the insurer with a loading of 0.3 (on the above) and assumes maintenance expenses allowance of 25% of the expected claim (assume continuous). The reinsurer has started the business with a capital of Rs 1.5 Cr.

- i) Calculate probability of ruin on second claim. (6)
- ii) Calculate probability of not becoming ruined within second year. (5)
- iii) If the reinsurance company wishes to keep maximum ruin probability of 10% at the end of fourth year then find out the minimum loading factor they should be charged on the expected claim amount. (4)

[15]

Q. 9) In a Cat Insurance policy, the number of claims (N) follow Poisson distribution with parameter λ . The claim amounts follow below distribution and are independent:

Claim amount	Probability
200	0.5p
300	0.3p
1000	1-0.8p

If the company has received 'd' number of claims each of size 300 then find out the distribution of $(N - d | d)$ and calculate its mean. [6]

Q. 10) η is the result of an integral equation as given below:

$$\eta = \int_0^1 (xe^x - 1)dx$$

- i) Derive the monte carlo estimator of the above using random numbers from uniform distribution $U(0, 1)$. (1)
- ii) Find out the minimum number of random numbers required for the above estimator to have an absolute error less than 0.2 with probability 95%. (6)

[7]
